Algorithm 1 Meta-evolution

Inputs: baseline; G_{total} total generations; G_{simplify} simplification generations; O offspring; P population size; training tasks T; validation tasks T_{val}; F fitness metric; N_{init} initial evaluation seeds; N_{val} validation seeds; reevaluation `strategy`: either `TTTS` with B seeds per generation, or `uniform` with N_{reeval} extra seeds per new population member.

`population` $\leftarrow$ [`baseline`]
`archive` $\leftarrow \{\}$
`new_members` $\leftarrow \{\}$
for $i = 1, \ldots, G_{\text{total}}$ **do**
  `offspring` $\leftarrow \{\}$ ▷ *Generate offspring*
  for $o = 1, \ldots, O$ **do**
    `sr_operator` $\leftarrow$ RandomChoice([`mutation`, `survival`, `selection`, `loss`])
    if $i = 1$ **then**
      `mode` $\leftarrow$ `explore`
    else if $G_{\text{total}} - i < G_{\text{simplify}}$ **then**
      `mode` $\leftarrow$ `simplify`
    else
      `mode` $\leftarrow$ RandomChoice([`explore`, `refine`, `crossover`, `simplify`])
    end if
    `parent` $\leftarrow$ TournamentSelection(`population`, $n = 2$)
    if `mode` $=$ `crossover` **then**
      `parent2` $\leftarrow$ TournamentSelection(`population`, $n = 2$)
    end if
    `feedback` $\leftarrow$ GenerateTraceFeedback(T, `parent`)
    `prompt` $\leftarrow$ GeneratePrompt(`sr_operator`, `mode`, `parent`, `feedback`, `parent2`)
    repeat
      `x` $\leftarrow$ `LLM_ensemble.sample().sample_completion(prompt)`
    until PassesValidation(`x`)
    `offspring` $\leftarrow$ `offspring` $\cup \{$`x`$\}$
  end for
  if `strategy` $=$ `TTTS` **then** ▷ *Choose reevaluations*
    $\psi_x \leftarrow$ top-two Thompson probability of x under $\mathcal{N}(\hat{\mu}_x, \hat{\sigma}^2/n_x)$, for $x \in$ `archive`
    `reeval` $\leftarrow B$ draws from `archive` with probability $\propto \psi_x$
  else
    `reeval` $\leftarrow \{x : N_{\text{reeval}} \mid x \in$ `new_members`$\}$
  end if
  `evaluation_batch` $\leftarrow \{x : N_{\text{init}} \mid x \in$ `offspring`$\} \cup$ `reeval` ▷ *Parallel evaluation*
  ParallelEvaluation(`evaluation_batch`, T, F)
  for $x \in$ `evaluation_batch` **do** ▷ *Update archive of scores*
    $\hat{\mu}_x \leftarrow$ mean of $F(x, \text{task}, \text{seed})$ over all evaluated tasks and seeds
    $n_x \leftarrow$ number of seeds evaluated
    `archive` $\leftarrow$ `archive` $\cup \{x\}$
  end for
  if $G_{\text{total}} - i < G_{\text{simplify}}$ **then** ▷ *LOC versus score population*
    Form the Pareto frontier of score versus LOC.
    Fill remaining slots (up to P) with individuals selected by highest overall score.
  else ▷ *Task-based population*
    For each task, add the individual with highest average score on that task to population.
    Fill remaining slots (up to P) with individuals selected by highest overall score.
  end if
  `new_members` $\leftarrow \{x \in$ `offspring` $\mid x \in$ `population`$\}$
  $x_i^* \leftarrow \arg\max_{x \in \texttt{population}} \hat{\mu}_x$ ▷ *Validate generation's best*
  `val`$[x_i^*] \leftarrow \frac{1}{N_{\text{val}}|T_{\text{val}}|} \sum_{\text{seeds, task} \in T_{\text{val}}} F(x_i^*, \text{task}, \text{seed})$
end for
`best_individual` $\leftarrow \arg\max_x$ `val`$[x]$
return `best_individual`